Predictive Maintenance System Analysis Report

Overview

This report presents the system analysis for a **Predictive Maintenance System Using Machine Learning**. The proposed system predicts whether a machine will require maintenance within the next 7 days using sensor readings such as temperature, vibration, and pressure, and it is intended to reduce unplanned downtime and improve maintenance decision-making.

The project planning document defines the project as an ML-based predictive maintenance solution with objectives centered on reliable maintenance prediction, early warning generation, and data-driven maintenance scheduling. The requirements document further identifies the main stakeholders, system use cases, functional requirements, and non-functional constraints that shape the system analysis.

Problem Statement

Industrial maintenance is often performed either reactively after failure or periodically without using actual machine condition data, which can result in unnecessary maintenance cost, unexpected downtime, and lower operational efficiency. A predictive maintenance system addresses this gap by analyzing machine condition signals and estimating whether intervention is needed before failure occurs.

The proposed system focuses on software-based predictive support rather than direct physical maintenance execution. Its purpose is to transform sensor and historical data into actionable maintenance predictions that help technicians and operators respond earlier and more effectively.

Project Objectives

The project has three primary objectives defined in the proposal and project plan:

- Build a reliable binary classification model for maintenance prediction.
- Provide early warnings to reduce unexpected machine downtime.
- Enable data-driven maintenance scheduling using prediction outputs.

The requirements and literature review also imply an additional technical objective: the system should support model improvement over time through retraining, logging, and maintainable architecture.

Dataset Analysis

The dataset description indicates that the project uses a large-scale synthetic industrial dataset designed for predictive maintenance, anomaly detection, and industrial machine learning use cases. The dataset includes records from 500,000 simulated machines across more than 30 machine types and contains both common sensor fields and machine-specific operational attributes.

The dataset contains core signals such as temperature, vibration, sound, power, and oil or coolant levels, in addition to maintenance history and AI supervision fields. It also provides two target columns: **Remaining_Useful_Life_days** for regression and **Failure_Within_7_Days** for binary classification. Since the proposal emphasizes prediction of maintenance within the next 7 days, the primary system analysis should treat **Failure_Within_7_Days** as the main target while presenting remaining useful life estimation as an extension or future enhancement.

The use of synthetic data is also important from a design perspective because it means the project does not depend on direct integration with real industrial hardware in its first version. This aligns with the documented scope, which excludes hardware integration and physical execution of maintenance.

Scope Definition

The project scope is clearly stated in the proposal and planning documents. The system covers data collection from sensors and prediction output, but it does not cover physical maintenance execution or hardware integration.

In Scope

- Collecting or receiving machine sensor data such as temperature, vibration, and pressure.
- Preprocessing data and preparing features for model inference.
- Predicting whether maintenance is needed within the next 7 days.
- Displaying results in a simple way for end users and maintenance staff.
- Logging predictions, model versions, and input records for monitoring and retraining.
- Supporting model updates and system administration functions.

Out of Scope

- Physical repair operations and maintenance execution.
- Deep industrial hardware or PLC integration in the initial version.
- Spare-part inventory and advanced enterprise maintenance planning modules.

Stakeholders

The requirements document identifies four core stakeholder groups.

Stakeholder	Role in the System	Main Need
End Users	Enter sensor readings and request predictions.	Simple prediction results and easy interaction.
Maintenance Technicians	Act on generated alerts and plan maintenance actions.	Clear status and decision support.
ML Engineers	Build, evaluate, retrain, and update the model.	Access to data, metrics, and model lifecycle support.
System Administrators	Maintain deployments, uptime, and platform health.	Monitoring, access control, and system reliability.

The project planning document also identifies development responsibilities across ML, backend, frontend, data engineering, and project management roles, which supports a modular analysis and architecture approach.

Functional Requirements

The system shall satisfy the following main functional requirements derived from the requirements and planning files:

1. The system shall accept machine sensor inputs including temperature, vibration, pressure, and other relevant features.
2. The system shall validate input data before passing it to the trained model.
3. The system shall run a trained ML classification model to predict whether maintenance is needed within the next 7 days.
4. The system shall return a binary prediction result and may display a confidence score if available.
5. The system shall store prediction requests, results, timestamps, and model version information.
6. The system shall support model updating and retraining by authorized users.
7. The system shall provide prediction history and monitoring views where needed.
8. The system shall present prediction output in a simple and understandable form for non-technical users.

Non-Functional Requirements

The requirements documents define a set of non-functional expectations for the platform.

Category	Requirement
Performance	Prediction response time should be under 2 seconds.
Availability	System uptime should be at least 99 percent.

Category	Requirement
Scalability	The system should handle multiple prediction requests.
Security	Sensor data should be stored and transmitted securely.
Maintainability	The model and codebase should be easy to update.
Usability	Outputs should be understandable for non-technical users.

The project plan also identifies risks such as imbalanced data, overfitting, noisy or missing sensor values, and deployment failure, which should be treated as quality and reliability constraints during system design.

Business Rules and Assumptions

Several business and design assumptions can be derived from the project documents and dataset description.

- A machine can be classified as either likely to require maintenance within 7 days or not likely to require maintenance within 7 days.
- Sensor readings must be available in structured digital form before inference is performed.
- Prediction results are intended to support maintenance decisions rather than automatically trigger machine shutdown or repair actions.
- Synthetic data is acceptable for analysis, training, validation, and academic demonstration in the current project phase.
- Future extensions may include regression-based remaining useful life prediction, anomaly detection, or tighter IoT integration.

System Constraints

The project operates under academic, technical, and scope constraints defined across the supplied materials.

- The project timeline requires completion of system analysis and design before implementation and final presentation milestones.
- The initial system should remain explainable and practical for academic evaluation, which supports choosing a classification-first design.
- Data quality issues such as imbalance, missing values, and noise must be addressed in preprocessing and evaluation.
- The system should prioritize maintainability and clear architecture over overly complex deployment choices in the first version.

Feasibility Analysis

Technical Feasibility

The project is technically feasible because the dataset is already defined, the required problem is well bounded, and the literature review supports classification-based predictive maintenance as a practical and interpretable starting point. The required system components such as preprocessing, classification, prediction API, and dashboard are also consistent with standard ML application design patterns.

Operational Feasibility

The system is operationally feasible because the identified users have clear interactions with the platform: operators provide readings, technicians act on alerts, ML engineers maintain models, and administrators monitor service health. The expected outputs are simple enough to support actual maintenance decisions in an academic or prototype industrial setting.

Schedule Feasibility

The project plan includes explicit phases for proposal, literature review, requirements gathering, system analysis and design, implementation, and final presentation, which indicates an organized development path. Because the project scope intentionally excludes physical maintenance execution and hardware integration, the schedule is more realistic and manageable for a student-scale system.

Risks

The planning document identifies several risks that are directly relevant to the system analysis.

Risk	Likelihood / Impact	Mitigation
Imbalanced dataset with few maintenance cases	High / High.	Apply oversampling such as SMOTE and evaluate recall carefully.
Model overfitting	Medium / High.	Use cross-validation and regularization.
Missing or noisy sensor data	Medium / High.	Apply validation, cleaning, and imputation in preprocessing.
Deployment failure	Low / High.	Test in a staging-like environment before release.
Limited explainability	Medium / Medium from literature concerns.	Use interpretable baselines and explain key features.

Analysis Conclusion

The project is well defined as a machine-learning-driven predictive maintenance decision-support system built around short-horizon failure prediction using structured sensor data. Its strongest analytical characteristics are a clearly bounded scope, measurable performance goals, realistic stakeholder roles, and alignment with literature that supports classification-based predictive maintenance for academic and practical use.

The system analysis also shows that the project should be designed as a modular platform with clear separation between data input, preprocessing, model inference, result presentation, and model lifecycle management. This makes the project suitable for the next design stages, including system architecture, database design, UML diagrams, DFDs, and deployment modeling.